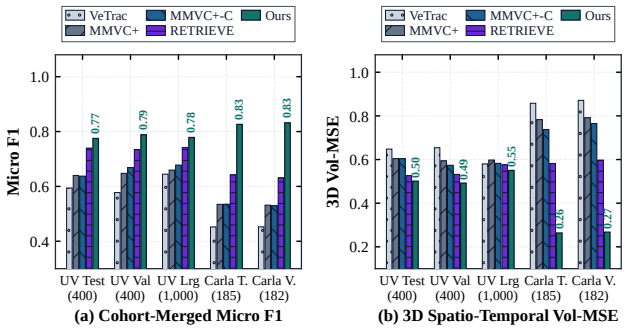
Supplementary Material: Trajectory Recovery via Constrained LLM Reasoning and Difficulty-Aware Routing



(a) Cohort-Merged Micro F1 **(b) 3D Spatio-Temporal Vol-MSE**

Figure 1: Multi-split benchmark comparison complementing Figure 5a of the main paper: (a) Cohort-Merged Micro F1 and (b) 3D Spatio-Temporal Volume MSE (Vol-MSE) across all five standard splits on UrbanVehicle and plate-free Carla.

A EXTENDED MULTI-METRIC BENCHMARKS AND METRIC PROPERTIES

Section 4.2 (Table 1) of the main paper reports end-to-end recovery performance across the five standard held-out splits on real-world UrbanVehicle (UrbanVehicle Test $N = 400$, UrbanVehicle Val $N = 400$, and UrbanVehicle Large-Scale $N = 1{,}000$) and high-density plate-free Carla (Carla Test $N = 185$ and Carla Val $N = 182$). This appendix details our cross-method evaluation protocol, reports the complete end-to-end benchmark ledger including deterministic anchor controls across all five splits (Figure 1 and Table 1), and proves the timestamp-projection invariance property of LCSS under uncalibrated temporal arrival drift.

A.1 Cross-Method Evaluation Protocol

To ensure an aligned comparison across VeTrac, MMVC+, MMVC+-C, RETRIEVE, deterministic graph controls (SigCluster + Nearest-Node and SigCluster + Unconstrained Dijkstra), and our C1–C3 recovery framework (**Ours**), our unified evaluation pipeline enforces three strict protocol invariants:

(1) **Identical ground-truth vehicle cohorts.** Across every held-out split on UrbanVehicle (UrbanVehicle Test $N = 400$, UrbanVehicle Val $N = 400$, UrbanVehicle Large-Scale $N = 1{,}000$, and complete held-out triggered cohorts Test48 $N = 48$ and Val37 $N = 37$, with routers and rankers trained on source-disjoint Train32 $N = 32$ and Train12 $N = 12$) and plate-free Carla (Carla Test $N = 185$ and Carla Val $N = 182$), all methods are evaluated against the *exact same ground-truth trajectory subset* $\mathcal{T}^*_{\text{sub}}$. For methods that decouple multi-vehicle visual clusters into sub-trajectories (RETRIEVE's refined 3D voxel sub-clusters and C2's decoupled corridor tracks on plate-free Carla), each reconstructed track is matched to its dominant ground-truth vehicle under the same $q(g)$ rule (main paper Section 2.3) so multi-vehicle clusters are scored without cross-ID truncation.

(2) **Unified five-metric accuracy and efficiency suite.** Every method is scored via identical metric definitions (Section 2.3 of the main paper): (i) per-vehicle Macro F1; (ii) cohort-merged Micro F1; (iii) bidirectional Spatio-Temporal Linear Combine (STLC, $\lambda = 0.5$); (iv) normalized Longest Common Subsequence (LCSS); and (v) 3D Spatio-Temporal Volume MSE (Vol-MSE over a 128^3 spatiotemporal voxel grid), alongside per-query latency, token volume, API cost, and LLM invocation rate.

(3) **Complete camera-to-road projection and temporal alignment on Carla.** On Carla (20 fps simulation clock across 1,169 cameras: 426 intersection and 743 mid-block cameras on a 426-node road network), all methods operate on second-aligned timestamps ($t \in [0, 86{,}400]$ s) and the complete 1,169-camera Haversine road-node projection table, evaluated over the exact same $N = 182$ validation and $N = 185$ test ground-truth vehicle trajectories without cross-ID truncation. Because camera density on Carla covers every road node (1,169 cameras vs. 426 nodes, unlike sparse UrbanVehicle with 157–198 cameras across 8,270 nodes), unexpanded *SigCluster + Nearest-Node* (0.4732 / 0.4528 Macro F1) already observes most traversed intersections, whereas unconditional shortest-path chaining (*SigCluster + Unconstrained Dijkstra*, 0.1581 / 0.1460 Macro F1 in Table 1) connects co-clustered visual outliers and opposite-lane mid-block projections into long multi-hop detours of false intermediate road nodes across intersections.

A.2 Why Timestamp-Agnostic LCSS Is Insensitive to Temporal Arrival Drift

Given a ground-truth trajectory $T^*_g = \{(u^*_{g,j}, [t^{\text{start}}_{g,j}, t^{\text{end}}_{g,j}])\}_{j=1}^{|T^*_g|}$ and its paired predicted trajectory $\hat{T}_q = \{(u_l, \hat{t}_l)\}_{l=1}^{|\hat{T}_q|}$ for $q = q(g)$:

(1) **Time-windowed node matching (Macro F1 and Micro F1):** A predicted point $(u_l, \hat{t}_l)$ counts as a hit H_g if and only if $u_l = u^*_{g,j}$ and $t^{\text{start}}_{g,j} \le \hat{t}_l \le t^{\text{end}}_{g,j}$. Per-vehicle precision $P_g = H_g/|\hat{T}_q|$ (0 when $|\hat{T}_q| = 0$) and recall $R_g = H_g/|T^*_g|$ (as well as 3D Spatio-Temporal Volume MSE) penalize both false detour nodes and any interpolated node whose timestamp $\hat{t}_l$ misses the true arrival window $[t^{\text{start}}_{g,j}, t^{\text{end}}_{g,j}]$.

(2) **Timestamp-projection invariance of LCSS under temporal drift:** In the standard evaluation pipeline, LCSS is defined as the length of the longest common node subsequence normalized by $\max(|\hat{T}_q|, |T^*_g|)$:

$$\text{LCSS}(\hat{T}_q, T^*_g) = \frac{\left|\text{LCS}\left(\{u_l\}_{l=1}^{|\hat{T}_q|},\ \{u^*_{g,j}\}_{j=1}^{|T^*_g|}\right)\right|}{\max\left(|\hat{T}_q|,\ |T^*_g|\right)}. \tag{1}$$

Because $\text{LCSS}(\hat{T}_q, T^*_g)$ projects out all arrival timestamps $\hat{t}_l$, any trajectory $\hat{T}'_q = \{(u_l, \hat{t}_l + \Delta t_l)\}$ that traverses the corridor nodes in topological order under uncalibrated signal-delay or travel-time interpolation achieves $\text{LCSS}(\hat{T}'_q, T^*_g) = 1.0$ (for example, on UrbanVehicle Test vehicle `vid=35613`, MMVC+-C matches

Table 1: Complete end-to-end comparison across the five standard held-out splits on UrbanVehicle (Test $N = 400$, Val $N = 400$, Large-Scale $N = 1{,}000$) and plate-free Carla (Test $N = 185$ and Val $N = 182$) under the unified five-metric suite, including deterministic anchor controls, absolute Macro F1 gains (ΔMacro F1), and relative 3D Vol-MSE reductions.

Dataset & Split (N)	Method	Macro F1 ↑	Micro F1 ↑	STLC ↑	LCSS ↑	Vol-MSE ↓	Gain ΔMacro F1	Rel. ΔVol-MSE
UrbanVehicle Test ($N = 400$)	VeTrac	0.4501	0.5942	0.6902	0.4512	0.6479	+0.3014	−22.7%
	MMVC+	0.6025	0.6400	0.8495	0.7207	0.6042	+0.1490	−17.1%
	MMVC+-C	0.6017	0.6376	0.8413	0.7308	0.6037	+0.1498	−17.0%
	RETRIEVE	0.7164	0.7398	0.8552	0.6814	0.5263	+0.0351	−4.8%
	SigCluster + Nearest-Node (w/o Expansion)	0.4050	0.3885	0.6872	0.3083	0.6614	+0.3465	−24.3%
	SigCluster + Unconstrained Dijkstra (Tier 1)	0.6123	0.6676	0.8161	0.6155	0.5195	+0.1392	−3.6%
	Ours (C1–C3)	**0.7515**	**0.7749**	**0.8793**	**0.7490**	**0.5010**	**Best**	**Best**
UrbanVehicle Val ($N = 400$)	VeTrac	0.4373	0.5775	0.6807	0.4453	0.6542	+0.3313	−24.8%
	MMVC+	0.5991	0.6478	0.8482	0.7220	0.5939	+0.1695	−17.2%
	MMVC+-C	0.6347	0.6688	0.8654	**0.7660**	0.5738	+0.1339	−14.3%
	RETRIEVE	0.7136	0.7340	0.8550	0.6812	0.5322	+0.0550	−7.6%
	SigCluster + Nearest-Node (w/o Expansion)	0.4133	0.3878	0.6938	0.3155	0.6638	+0.3553	−25.9%
	SigCluster + Unconstrained Dijkstra (Tier 1)	0.6332	0.6912	0.8297	0.6356	0.5037	+0.1354	−2.4%
	Ours (C1–C3)	**0.7686**	**0.7884**	**0.8908**	0.7643	**0.4919**	**Best**	**Best**
UrbanVehicle Large-Scale ($N = 1{,}000$)	VeTrac	0.5532	0.6447	0.7656	0.5558	0.5797	+0.1969	−5.1%
	MMVC+	0.6159	0.6596	0.8405	0.6937	0.5979	+0.1342	−8.0%
	MMVC+-C	0.6469	0.6778	0.8623	**0.7380**	0.5827	+0.1032	−5.6%
	RETRIEVE	0.7178	0.7420	0.8522	0.6949	0.5754	+0.0323	−4.4%
	SigCluster + Nearest-Node (w/o Expansion)	0.3621	0.3468	0.6718	0.2758	0.6758	+0.3880	−18.6%
	SigCluster + Unconstrained Dijkstra (Tier 1)	0.5709	0.6246	0.7965	0.5539	0.5864	+0.1792	−6.2%
	Ours (C1–C3)	**0.7501**	**0.7785**	**0.8751**	**0.7360**	**0.5501**	**Best**	**Best**
Carla Test ($N = 185$)	VeTrac	0.4293	0.4521	0.7483	0.4976	0.8578	+0.3846	−69.3%
	MMVC+	0.5152	0.5347	0.7854	0.6056	0.7828	+0.2987	−66.3%
	MMVC+-C	0.5183	0.5352	0.7576	0.6232	0.7373	+0.2956	−64.2%
	RETRIEVE	0.6304	0.6417	0.8102	0.6310	0.5822	+0.1835	−54.7%
	SigCluster + Nearest-Node (w/o Expansion)	0.4732	0.4976	0.7740	0.5291	0.8462	+0.3407	−68.8%
	SigCluster + Unconstrained Dijkstra (Tier 1)	0.1581	0.1852	0.7205	0.1432	0.8321	+0.6558	−68.3%
	Ours (C1–C3)	**0.8139**	**0.8262**	**0.8919**	**0.7643**	**0.2636**	**Best**	**Best**
Carla Val ($N = 182$)	VeTrac	0.4239	0.4523	0.7428	0.5003	0.8712	+0.3954	−69.3%
	MMVC+	0.5072	0.5318	0.7773	0.6149	0.7914	+0.3121	−66.2%
	MMVC+-C	0.5028	0.5294	0.7601	0.6135	0.7651	+0.3165	−65.0%
	RETRIEVE	0.6110	0.6307	0.8101	0.6179	0.5976	+0.2083	−55.3%
	SigCluster + Nearest-Node (w/o Expansion)	0.4528	0.4820	0.7670	0.5162	0.8615	+0.3665	−69.0%
	SigCluster + Unconstrained Dijkstra (Tier 1)	0.1460	0.1735	0.7174	0.1365	0.8446	+0.6733	−68.3%
	Ours (C1–C3)	**0.8193**	**0.8320**	**0.8972**	**0.7639**	**0.2674**	**Best**	**Best**

all 10/10 road nodes in sequence for LCSS = 1.000, yet 9 of 10 interpolated timestamps miss $[t^{\text{start}}_{g,j}, t^{\text{end}}_{g,j}]$, yielding $H_g = 1$ and $\text{F1}_g = 0.100$). Consequently, while **Ours** surpasses MMVC+-C on LCSS on UrbanVehicle Test and both Carla splits, MMVC+-C records a marginally higher untimed LCSS (+0.0017–0.0020) on UrbanVehicle Val and Large-Scale while trailing **Ours** by 10.3–13.4 percentage points in Macro F1 and 5.6%–14.3% in 3D Vol-MSE on those two splits (Table 1).

B THEORETICAL PROOFS

B.1 Proof of Proposition 1 (Boundary Clamping and Zero-Leakage Abstention)

Proof. Let $O_q = (o_1, \ldots, o_m)$ be a triggered candidate cluster ($\gamma_{\text{inv}}(O_q) = 1$) with candidate local sub-interval bindings $\mathcal{B}_{\text{local}}$ retrieved from the source-disjoint bank $\mathcal{E}_{\text{bank}}$ ($s_e \neq s_q$). Consider any binding $b \in \mathcal{B}_{\text{local}}$ overriding camera c_{i_b} from its Top-1 prior $v_{\text{pub}} = v_{c_{i_b},1}$ to secondary approach $v_{\text{exp}} = v_{c_{i_b},2}$:

(1) *Bidirectional clamping at interior observations ($\text{bnd}(b) = 0$):* When $1 < i_b < m$, node $v^*_{i_b}$ is simultaneously constrained by the upstream interval $P_{\text{SP}}(v^*_{i_b-1}, v^*_{i_b})$ and the downstream interval $P_{\text{SP}}(v^*_{i_b}, v^*_{i_b+1})$. Under C2's bounded Dijkstra execution, replacing v_{pub} with v_{exp} must preserve directed reachability and kinematic feasibility ($\hat{v} \le v_{\max}$) on both adjacent intervals, preventing unconstrained endpoint loss.

(2) *Open-boundary single-source truncation:* When $\text{bnd}(b) = 1$ and $\text{supp}(b) = 1$ at $i_b = 1$ (or $i_b = m$), the outer side of c_{i_b} has no clamping anchor in O_q. If a single-source local match ($k = 1$) maps c_1 to a downstream approach v_{exp} satisfying $P_{\text{SP}}(v_{\text{exp}}, v^*_2) \subsetneq P_{\text{SP}}(v_{\text{pub}}, v^*_2)$ (or maps c_m to an upstream approach v_{exp} with $d_{\text{SP}}(v^*_{m-1}, v_{\text{exp}}) < d_{\text{SP}}(v^*_{m-1}, v_{\text{pub}})$), C2 drops all links in $P_{\text{SP}}(v_{\text{pub}}, v_{\text{exp}})$, strictly reducing recall R_g while leaving internal nodes unchanged ($\Delta\text{F1}_g < 0$). Because $\text{Gate}_{G_5}(\mathcal{B}_{\text{local}})$ requires $\exists\, b \in \mathcal{B}_{\text{local}}$ with $\text{bnd}(b) = 0 \vee \text{supp}(b) \ge 2$, every query whose candidate set $\mathcal{B}_{\text{local}}$ contains only single-source open-boundary overrides is deterministically rejected ($\mathcal{R}_q \leftarrow \varnothing$), guaranteeing $\hat{T}_q(G_5) \equiv \hat{T}_q(\text{No-Ret})$ and $\Delta\text{F1}_g = 0$ across all unconstrained single-source boundary queries.

(3) *Terminal signal-delay separation under G_5^{kin}:* For $\text{supp}(b) \ge 2$ at $i_b = m$ with $\text{trunc}_m(b) = 1$, pass-through teacher intervals $e \in \mathcal{E}_{\text{pass}}(b)$ ($c^{\text{teacher}}_{\text{next}} \neq \varnothing$) traversed $v_{\text{prev}} \to v_{\text{exp}}$ ($v_{\text{prev}} = v_{c_{m-1},1}$) before continuing past the intersection. A query vehicle terminating at c_m validly stops at the upstream queue node v_{exp} (rather than exiting via v_{pub}) if and only if its observed interval duration $\Delta t_{m-1\to m} = t_m - t_{m-1}$ includes red-light waiting time exceeding the teachers' pass-through duration $\max_{e\in\mathcal{E}_{\text{pass}}(b)} \Delta t_{\text{teacher}}(e)$ and satisfies $d_{\text{SP}}(v_{\text{prev}}, v_{\text{exp}})/\max(\Delta t_{m-1\to m}, \epsilon) \le$

$v_{\text{stop}} = 5.0$ m/s. Both quantities are computed directly from pre-call inputs $(O_q, \mathcal{G}, \mathcal{E}_{\text{bank}})$ in $O(|\mathcal{B}_{\text{local}}|)$ time (< 1 ms) without ground-truth leakage or extra LLM calls. □

B.2 Proof of Proposition 2 (Subgraph Equivalence and Topological Validity)

Proof. Fix a deterministic tie-breaking rule on equal-length shortest paths in $\mathcal{G}$ (lexicographic node-ID order, matching C2's bounded Dijkstra priority queue). By Definition 1 and Eq. (8) of the main paper, for every index pair $(a, b) \in \mathcal{I}_q = \{(i, i+1)\}_{i=1}^{m-1} \cup \{(i-1, i+1)\}_{i=2}^{m-1}$ and every candidate approach pair $(u, v) \in \Gamma_2(c_a) \times \Gamma_2(c_b)$, all vertices $V_{\text{SP}}(u, v)$ (alongside candidate via-nodes $\mathcal{W}_q$) and all induced directed edges $\mathcal{E}_q = \mathcal{E} \cap (\mathcal{V}_q \times \mathcal{V}_q) \supseteq E_{\text{SP}}(u, v)$ along $P_{\text{SP}}^{\mathcal{G}}(u, v)$ are included in $\mathcal{G}_q = (\mathcal{V}_q, \mathcal{E}_q) \subseteq \mathcal{G}$. Because edge weights $\ell(e) > 0$ are identical on $\mathcal{E}_q \subseteq \mathcal{E}$ and no directed path in $\mathcal{G}_q \subseteq \mathcal{G}$ can have length strictly smaller than $d_{\text{SP}}^{\mathcal{G}}(u, v)$, deterministic tie-breaking yields $d_{\text{SP}}^{\mathcal{G}_q}(u, v) = d_{\text{SP}}^{\mathcal{G}}(u, v)$ and exact path equality $P_{\text{SP}}^{\mathcal{G}_q}(u, v) = P_{\text{SP}}^{\mathcal{G}}(u, v)$ for all $(u, v) \in \Gamma_2(c_a) \times \Gamma_2(c_b)$.

Next, consider C2's three-tier execution on $\mathcal{G}_q$ for any planned interval $(u \to \tilde{w}_{i,1} \to \tilde{w}_{i,2} \to v)$ with $(u, v) \in \Gamma_2(c_a) \times \Gamma_2(c_b)$: (i) Under Tier 1, if the proposed via-nodes lie in $\mathcal{V}_q$ and directed paths $P_{\text{SP}}^{\mathcal{G}_q}(u, \tilde{w}_{i,1})$, $P_{\text{SP}}^{\mathcal{G}_q}(\tilde{w}_{i,1}, \tilde{w}_{i,2})$, and $P_{\text{SP}}^{\mathcal{G}_q}(\tilde{w}_{i,2}, v)$ exist and satisfy C2's speed, span, and zero-detour bounds, every edge in the concatenated path belongs to $\mathcal{E}_q \subseteq \mathcal{E}$ (moreover, whenever $\tilde{w} \in V_{\text{SP}}(u, v)$, optimal substructure guarantees that concatenating $P_{\text{SP}}^{\mathcal{G}_q}(u, \tilde{w})$ and $P_{\text{SP}}^{\mathcal{G}_q}(\tilde{w}, v)$ recovers $P_{\text{SP}}^{\mathcal{G}}(u, v)$). (ii) If any proposed via-node $\tilde{w}_{i,k}$ is outside $\mathcal{V}_q$, unreachable in $\mathcal{G}_q$, or violates kinematic/detour bounds, C2 deterministically drops $\tilde{w}_{i,k}$ and falls back to Tier 2's direct shortest path $P_{\text{SP}}^{\mathcal{G}_q}(u, v) = P_{\text{SP}}^{\mathcal{G}}(u, v) \subseteq \mathcal{E}_q$. (iii) If (v_i^*, v_{i+1}^*) itself violates kinematic or mid-block zero-detour bounds, C2 trims isolated boundary spikes ($i \in \{1, m\}$) or invokes Tier 3 on interior $(i-1, i+1) \in \mathcal{I}_q$, expanding $P_{\text{SP}}^{\mathcal{G}_q}(v_{i-1}^*, v_{i+1}^*) = P_{\text{SP}}^{\mathcal{G}}(v_{i-1}^*, v_{i+1}^*) \subseteq \mathcal{E}_q$. Thus every emitted interval sequence $(w_{i,0}, \ldots, w_{i,L_i})$ is a valid directed path in $\mathcal{E}_q \subseteq \mathcal{E}$, and boundary anchor deduplication $(w_{i,L_i} = w_{i+1,0})$ preserves directed adjacency across interval joins, guaranteeing $(u_l, u_{l+1}) \in \mathcal{E}$ for all $1 \le l < |\hat{T}_q|$ and ensuring 100% road-network validity (0% segment violations). □

C C2 PROMPT CONTRACT, TIER ABLATION, AND MULTI-SEED OUTLIER ROBUSTNESS

C.1 Prompt and JSON Schema Contracts

Each triggered cluster O_q is serialized into a structured prompt containing `[CLUSTER_TRACK_FACTS]` (ordered observation table, Top-2 candidate road nodes $\Gamma(c_i)$, output contract, and corridor subgraph $\mathcal{G}_q$), `[PAIR_COMPARISON_FACTS]` (directed shortest-path lengths, implied speeds, and candidate intermediate intersections), and (when accepted by G_5/G_5^{kin}) `[MAPPING_EXPERIENCE_JSON]` (≤ 4,000 chars). The LLM outputs a single JSON object adhering to the C2 road-intent schema under deterministic greedy decoding (temperature 0.0, random seed 0). Across the complete held-out Test48 cohort ($N = 48$, Table 3 of the main paper), decoupling structured road-intent planning from C2's bounded Dijkstra verifier reduces disconnected output segments from 88.5% (108/122 segments under Direct-LLM and Prompt-Rules) to 0.0% (0/699 under C2 Only w/o C1 and 0/519 under Full C1–C3, Routed), lifting Macro F1 from 0.3507 to 0.7062 (C2 Only w/o C1, 37W/4T/7L) and **0.8540** (Full C1–C3, Routed, 47W/0T/1L):

```
{
"camera_node_choices": [{"camera_id": 33, "road_node": 281}],
"tracks": [{"obs_ids": ["record:242216", "record:242217"],
"interval_intents": [{"from_obs_id": "record:242216",
"to_obs_id": "record:242217", "via_nodes": [6232]}]}],
"unassigned_obs_ids": []
}
```

C.2 Three-Tier Verification Ablation and Multi-Seed Outlier Robustness

Tables 2 and 3 (complementing Figure 7b of the main paper) evaluate (i) C2's three deterministic verification tiers across the full UrbanVehicle Val ($N = 400$) and UrbanVehicle Test ($N = 400$) splits under the unified five-metric suite, and (ii) 5-seed multi-metric robustness (mean ± std) under synthetic false-positive camera outlier injection ($\rho_{\text{noise}} \in \{0\%, 5\%, 10\%, 20\%\}$):

(1) **Incremental gains across Tiers 1–3 and C1–C3 (Table 2):** Tier 1 (unconstrained shortest-path expansion) recovers intermediate road nodes missed by Nearest-Node anchors, lifting LCSS by +30.7 to +32.0 points and Macro F1 from 0.4133 → 0.6332 (Val) and 0.4050 → 0.6123 (Test), but connects cross-city detours when adjacent observations contain noisy jumps. Tier 2 (kinematic span $d_{\max} = 1{,}200$ m and speed $v_{\max} = 25$ m/s gating) prunes these infeasible links before concatenation, lifting Macro F1 by +11.66 points on Val (0.7498) and +11.15 points on Test (0.7238). Adding Tier 3's two-hop spike bypass reaches 0.7509 / 0.7258 Macro F1 at zero API cost, and coupling C2 with C1's gated transition retrieval and C3 routing (**Full C1–C3**) reaches **0.7686** on Val and **0.7515** on Test.

(2) **Monotonically widening Tier-3 advantage under multi-seed outlier contamination (Table 3):** On unperturbed UrbanVehicle Test ($\rho_{\text{noise}} = 0\%$), isolated single-point spikes occur in a minority of clusters (+0.20 F1 points from Tier 3). As false-positive camera snapshots are injected across 5 independent random seeds ($\rho_{\text{noise}} = 5\% \to 10\% \to 20\%$), rejecting an outlier via Tier 2 alone severs the trajectory into disconnected fragments at every corrupted observation, causing Tier 1+2 Macro F1 to drop by −2.64, −4.79, and −9.12 percentage points (0.6326 ± 0.0068 at $\rho_{\text{noise}} = 20\%$). Full C2's Tier-3 two-hop bypass ($v_{i-1}^* \to v_{i+1}^*$) bridges across each excised spike, widening Tier 3's margin over Tier 1+2 from +0.20 points (0%) to +2.58 (5%), +4.26 (10%), and +7.42 percentage points (20%), restricting the Macro F1 drop vs. clean to −1.90 points (0.7068 ± 0.0025).

D C3 ROUTER FEATURES, COLD-START SCALING, AND SUBGRAPH PRUNING

D.1 Pre-Call Router Features and Cold-Start Experience Scaling

Table 4 specifies the 18-dimensional pre-call feature vector $\mathbf{x}_q \in \mathbb{R}^{18}$ (9 transformed structural/kinematic attributes and 9 binary missingness flags indicating absent retrieval candidates under $\mathcal{R}_q = \varnothing$ or single-interval edge cases) extracted in $O(m)$ time (< 1 ms)

Table 2: Incremental ablation of C2's three-tier corridor verification and full C1–C3 integration across UrbanVehicle Val ($N = 400$) and UrbanVehicle Test ($N = 400$) evaluated under the unified five-metric trajectory recovery suite.

Evaluation Split (N)	Verification Tier / Recovery Configuration	Macro F1 ↑	Micro F1 ↑	STLC ↑	LCSS ↑	Vol-MSE ↓
UrbanVehicle Val ($N = 400$)	w/o C2 (Nearest-Node Anchors Only)	0.4133	0.3878	0.6938	0.3155	0.6638
	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.6332	0.6912	0.8297	0.6356	0.5037
	C2 Tier 1+2 (+ Kinematic Span & Speed Gate)	0.7498	0.7693	0.8844	0.7474	0.5003
	Deterministic C2 Tier 1+2+3 (Zero-LLM, w/o C1)	0.7509	0.7715	0.8852	0.7489	0.4997
	Full (C1–C3)	**0.7686**	**0.7884**	**0.8908**	**0.7643**	**0.4919**
UrbanVehicle Test ($N = 400$)	w/o C2 (Nearest-Node Anchors Only)	0.4050	0.3885	0.6872	0.3083	0.6614
	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.6123	0.6676	0.8161	0.6155	0.5195
	C2 Tier 1+2 (+ Kinematic Span & Speed Gate)	0.7238	0.7473	0.8692	0.7230	0.5127
	Deterministic C2 Tier 1+2+3 (Zero-LLM, w/o C1)	0.7258	0.7500	0.8706	0.7248	0.5105
	Full (C1–C3)	**0.7515**	**0.7749**	**0.8793**	**0.7490**	**0.5010**

Table 3: Multi-seed robustness (mean ± std across 5 independent random seeds) under synthetic false-positive camera outlier contamination ($\rho_{\text{noise}} \in \{0\%, 5\%, 10\%, 20\%\}$) on UrbanVehicle Test ($N = 400$), alongside Macro F1 change (ΔMacro F1) relative to each variant's clean ($\rho_{\text{noise}} = 0\%$) baseline.

Outlier Contamination Rate	C2 Verification Variant	Macro F1 ↑	Micro F1 ↑	STLC ↑	LCSS ↑	ΔMacro F1 vs. Clean
$\rho_{\text{noise}} = 0\%$ (Clean Reference, UV Test $N = 400$)	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.6123 ± 0.0000	0.6676 ± 0.0000	0.8161 ± 0.0000	0.6155 ± 0.0000	0.00 pts (Clean Ref.)
	C2 Tier 1+2 (w/o Tier-3 2-Hop Spike Bypass)	0.7238 ± 0.0000	0.7473 ± 0.0000	0.8692 ± 0.0000	0.7230 ± 0.0000	0.00 pts (Clean Ref.)
	Deterministic Full C2 (Tier 1+2+3)	**0.7258 ± 0.0000**	**0.7500 ± 0.0000**	**0.8706 ± 0.0000**	**0.7248 ± 0.0000**	**0.00 pts (Clean Ref.)**
$\rho_{\text{noise}} = 5\%$ Contamination (UV Test $N = 400$, 5 Seeds)	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.5982 ± 0.0033	0.6575 ± 0.0028	0.8078 ± 0.0017	0.5953 ± 0.0035	−1.41 pts
	C2 Tier 1+2 (w/o Tier-3 2-Hop Spike Bypass)	0.6974 ± 0.0031	0.7239 ± 0.0026	0.8531 ± 0.0021	0.6899 ± 0.0036	−2.64 pts
	Deterministic Full C2 (Tier 1+2+3)	**0.7232 ± 0.0008**	**0.7475 ± 0.0006**	**0.8685 ± 0.0005**	**0.7220 ± 0.0008**	**−0.26 pts**
$\rho_{\text{noise}} = 10\%$ Contamination (UV Test $N = 400$, 5 Seeds)	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.5922 ± 0.0071	0.6529 ± 0.0029	0.8011 ± 0.0032	0.5832 ± 0.0067	−2.01 pts
	C2 Tier 1+2 (w/o Tier-3 2-Hop Spike Bypass)	0.6759 ± 0.0042	0.7049 ± 0.0025	0.8384 ± 0.0021	0.6618 ± 0.0053	−4.79 pts
	Deterministic Full C2 (Tier 1+2+3)	**0.7185 ± 0.0017**	**0.7429 ± 0.0014**	**0.8652 ± 0.0008**	**0.7159 ± 0.0019**	**−0.73 pts**
$\rho_{\text{noise}} = 20\%$ Contamination (UV Test $N = 400$, 5 Seeds)	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.5681 ± 0.0080	0.6320 ± 0.0038	0.7856 ± 0.0055	0.5502 ± 0.0098	−4.42 pts
	C2 Tier 1+2 (w/o Tier-3 2-Hop Spike Bypass)	0.6326 ± 0.0068	0.6650 ± 0.0058	0.8116 ± 0.0046	0.6083 ± 0.0091	−9.12 pts
	Deterministic Full C2 (Tier 1+2+3)	**0.7068 ± 0.0025**	**0.7309 ± 0.0030**	**0.8570 ± 0.0018**	**0.7014 ± 0.0033**	**−1.90 pts**

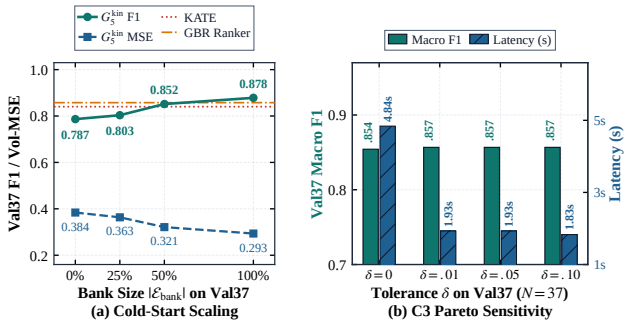


Figure 2: Cold-start experience-bank scaling and out-of-sample C3 Pareto routing sensitivity: (a) monotonic zero-loss Macro F1 and Vol-MSE improvements under G_5^{kin} on held-out Val37 ($N = 37$) as $|\mathcal{E}_{\text{bank}}|$ scales from 0% (cold-start C2 floor) to 100% (90 experiences), compared against KATE Cosine and GBR Ranker; (b) out-of-sample C3 quality–latency trade-off (Macro F1 on left axis; Mean Latency on right axis) across quality tolerance $\delta \in \{0.00, 0.01, 0.05, 0.10\}$ on held-out Val37 ($N = 37$, GBR routers trained on source-disjoint Train32).

before model dispatch. Figure 2a evaluates cold-start scaling on held-out Val37 ($N = 37$) as the experience bank $|\mathcal{E}_{\text{bank}}|$ grows from 0% (zero-shot C2 floor) through 25% (22 experiences) and 50% (45 experiences) to 100% (90 experiences). Because G_5^{kin} defaults cleanly to C2's verified zero-shot reasoning whenever a candidate experience lacks boundary support, Macro F1 rises monotonically ($0.7866 \rightarrow 0.8035 \rightarrow 0.8517 \rightarrow \mathbf{0.8784}$, with Vol-MSE dropping from 0.3840 to **0.2935**) with zero per-vehicle losses at every bank fraction (0L), whereas similarity/utility rankers (KATE and GBR) suffer 8–15 regressions by injecting unverified single-source or terminal free-flow overrides. Figure 2b confirms that relaxing C3's Pareto tolerance from $\delta = 0.00$ (4.84 s) to $\delta = 0.01$ (1.93 s) stabilizes latency and Macro F1 (0.8567) across $\delta \in [0.01, 0.10]$.

Table 4: Pre-call feature vector $\mathbf{x}_q \in \mathbb{R}^{18}$ (9 transformed structural/kinematic attributes +9 missingness flags) for the C3 router (Section 3.3.1 of the main paper).

Idx	Pre-Call Feature Attribute	Pre-Call Transform & Operational Definition
1, 10	`traj_len` (+ missing flag)	$\log(1 + \max(x, 0))$ of cluster observation count m
2, 11	`time_span` (+ missing flag)	$\log(1 + \max(x, 0))$ of cluster duration $t_m - t_1$ (s)
3, 12	`spatial_range` (+ missing flag)	$\log(1 + \max(x, 0))$ of camera coordinate bounding span
4, 13	`branching_score` (+ missing flag)	$\text{clip}(x, 0, 1)$ road-graph candidate branching ratio
5, 14	`missing_ratio` (+ missing flag)	$\text{clip}(x, 0, 1)$ estimated unobserved intermediate hop ratio
6, 15	`jump_distance` (+ missing flag)	$\log(1 + \max(x, 0))$ of max inter-camera jump distance
7, 16	`input_char_length` (+ missing flag)	$\log(1 + \max(x, 0))$ of serialized prompt character count
8, 17	`retrieval_count` (+ missing flag)	$\log(1 + \max(x, 0))$ of retrieved transition candidates
9, 18	`retrieval_similarity` (+ missing flag)	$\text{clip}(x, 0, 1)$ top retrieved structural similarity score

Table 5: Corridor subgraph pruning ($\mathcal{G} \to \mathcal{G}_q$, Eq. (8) of the main paper) graph compression across two-anchor interval prompts (164 cases, 547 intervals) and full multi-observation clusters ($N = 400$), shortest-path equivalence, and stage-wise non-LLM local CPU execution profile on UrbanVehicle Test.

Diagnostic Audit Scope	Graph Scope / Local Pipeline Stage	Mean Active Nodes	Mean Active Edges	Edge Reduction	Shortest-Path Parity	Execution Time (p50)
Two-Anchor Corridor Prompt (164 Cases, 547 Intervals)	Unpruned Full City Graph $\mathcal{G}$ (361.4 KB prompt)	8,270.0 (100.0%)	9,566.0 (100.0%)	0.00% (Ref.)	100.0% (547/547 Ref.)	13.53 s (Test48, $\mathcal{G}$)
	Pruned Corridor Subgraph $\mathcal{G}_q$ (21.9 KB prompt)	**34.9 (0.42%)**	**39.1 (0.41%)**	**−99.59%**	**100.0% (547/547 Exact)**	**1.98 s (Test48, Routed)**
Cluster-Wide $\mathcal{G}_q$ Pair Scope (UV Test $N = 400$ Multi-Obs.)	Consecutive Pairs Only: $\mathcal{I}_q = \{(i, i+1)\}$	88.0 (1.06%)	98.3 (1.03%)	−98.97%	Tier 1–2 Only (Misses 2-Hop)	0.65 ms (0.25 ms)
	Consecutive + 2-Hop: $\mathcal{I}_q$ (Default Eq. 8)	**99.6 (1.20%)**	**112.4 (1.18%)**	**−98.82%**	**100.0% (Full Tier 1–3)**	**0.71 ms (0.26 ms)**
	All Forward Pairs: $\mathcal{I}_q = \{(a, b) : a < b\}$	113.6 (1.37%)	129.4 (1.35%)	−98.65%	100.0% (Full Tier 1–3)	0.80 ms (0.25 ms)
Local Non-LLM CPU Profile (UV Test $N = 400$ Clusters)	Stage 1 Kinematic & G_5^{kin} Gate (γ_{inv})	–	–	–	Exact Pre-Call Filter	1.08 ms (0.26 ms)
	Stage 2 Corridor Subgraph Pruner ($\mathcal{G} \to \mathcal{G}_q$)	99.6 (1.20%)	112.4 (1.18%)	−98.82%	100.0% Exact SP Parity	0.71 ms (0.26 ms)
	Stage 3 Bounded C2 Dijkstra Verifier	99.6 (1.20%)	112.4 (1.18%)	−98.82%	100.0% Valid Edges	0.016 ms (0.010 ms)
	Total Non-LLM Local CPU (Stages 1–3)	–	–	–	**< 0.10% of Routed Latency**	**1.81 ms (0.53 ms)**

D.2 Corridor Subgraph Compression and Local Non-LLM CPU Profile

Table 5 audits (i) prompt-level corridor subgraph compression ($\mathcal{G} \to \mathcal{G}_q$, Eq. (8) of the main paper) across two-anchor interval queries (34.9 mean active nodes, 21.9 KB) and full multi-observation clusters on UrbanVehicle Test ($N = 400$, 99.6 mean active nodes across the union of consecutive and two-hop pairs), and (ii) stage-wise local non-LLM CPU execution overhead. Pruning the full city graph $\mathcal{G}$ (8,270 nodes, 9,566 directed edges) to $\mathcal{G}_q$ removes 98.8%–99.6% of nodes, 94.0% of prompt bytes, and 96.3%–97.4% of input tokens while preserving 100.0% (547/547) exact shortest-path equivalence.

Extending $\mathcal{I}_q$ from consecutive pairs $\{(i, i+1)\}$ (88.0 nodes) to include two-hop spike-bypass pairs $\{(i-1, i+1)\}$ (99.6 nodes) adds only 11.6 nodes and 0.06 ms of local CPU time (0.65 → 0.71 ms), yet guarantees that Tier-3 two-hop bypass paths remain inside $\mathcal{G}_q$ without computing all-pairs forward paths (113.6 nodes, 0.80 ms). Across Stages 1–3, local non-LLM CPU execution totals 1.81 ms per cluster (1.08 ms for Stage 1 γ_{inv} and G_5^{kin} gating, 0.71 ms for Stage 2 $\mathcal{G}_q$ construction, and 0.016 ms for Stage 3 C2 bounded Dijkstra verification), accounting for $< 0.10\%$ of the 1.98 s end-to-end routed service latency on held-out Test48.